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Quantitative text analysis – Essex Summer School

Dictionary methods and validation

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Today's class

- Dictionary methods and their validation
- Lab session: off-the-shelf dictionaries, custom dictionaries, dictionary development and validation.
- Breakout groups

Economic insecurity in House of Commons speeches

- Can we use a dictionary approach to measure when MPs talk about economic insecurity?
- Relevant speeches may mention wages, prices, jobs, inflation, bills, unemployment, poverty, or the cost of living
- That sounds simple, but consider the following:
 - should work count as an economic term?
 - does growth mean economic growth or something else?
 - what do we miss if MPs say families are struggling?

"... we have already introduced more Bills than Labour" last four Sessions in office by a considerable margin."



Penny Mordaunt, Conservative, **Business of the House**, 16 May 2024.

Dictionary methods

- “Dictionaries use the rate at which key words appear in a text to classify documents into categories or to measure the extent to which documents belong to particular categories” (Grimmer & Stewart, 2013)
 - That is, dictionaries **count** words associated with specific meanings
- Dictionaries consist of **key-value** pairs
 - **Key**: label for the concept or equivalence class
 - **Values**: (multiple) terms or patterns of terms that are declared equivalent occurrences of the key class

Dictionaries in Quanteda

A dictionary in **quanteda** assigns possible features to **categories**, or **"keys"**. It can have as many categories as you wish, and can be composed of any possible feature.

```
insecurity_dict <- dictionary(list(
  economic_insecurity = c(
    "inflation*", "price*", "wage*", "unemploy*",
    "poverty", "bills", "rent*", "debt*", "cost*"
  )
))

hoc_dfm <- corpus(hoc[1:3, ], text_field = "text") |>
  tokens(remove_punct = TRUE) |>
  dfm()

dfm_lookup(hoc_dfm, dictionary = insecurity_dict)
```

Documents have different lengths

Raw dictionary counts are not directly comparable when speeches differ in length, especially if we want to say something about the extent to which a document fits a category.

Speech	Length	Dictionary hits	Hits per 100 terms
short intervention	50	2	4.0
long speech	500	10	2.0

- The long speech has more hits, but the short speech is more densely about the concept
- We usually compare **rates**: dictionary hits divided by document length

What a first dictionary returns

Date	Agenda	Speaker	Hits	Rate per 100 terms	First read
1986-04-17	Unemployment Costs	Mr. Moore	4	8.7	likely relevant
2009-01-22	Energy and Climate Change	Lindsay Hoyle	4	7.7	likely relevant
2024-05-16	Business of the House	Penny Mordaunt	1	0.1	false positive
2025-05-01	Business of the House	Lucy Powell	3	0.4	mixed context

When adapting / developing a dictionary, it's the edge cases that require further attention.

Why dictionaries?

- Dictionary methods are a **deductive** strategy: the researcher defines the relevant words or phrases before counting
- This makes dictionary measures transparent and easy to inspect.
- But transparency $\neq$ validity: a dictionary can be clear, reproducible, and still measure the wrong thing

When are dictionaries useful? When are dictionaries really useful?

Useful when...

- the term list can be inspected and reported (don't use any dictionaries for which the content is proprietary)
- validation against human judgment is possible

Really useful when...

- The categories that we want to code are **manifest and concrete** rather than **latent and abstract**
 - Multifaceted theoretical constructs such as **frames** or **ideologies** are difficult to capture using dictionaries alone
- All **known synonyms** are included in the dictionary
- Dictionary entries do not have **multiple meanings**

The observations are adapted from Van Atteveldt *et al.* (2022).

Validating dictionaries: error analysis

Key questions when applying dictionaries concern their **validity**, **recall**, and **precision**.

- **Validity**: does the dictionary meaningfully operationalize our concept of interest?
- **Recall**: does the dictionary identify all relevant content?
- **Precision**: does the dictionary identify only relevant content?

Validating dictionaries: error types

	Human says relevant	Human says not relevant
Dictionary hits	true positives	false positives
Dictionary misses	false negatives	true negatives

- False positives lower **precision**
- False negatives lower **recall**

A small validation example

Suppose we hand-code 20 speeches after applying the dictionary.

	Human relevant	Human not relevant
Dictionary hits	6	2
Dictionary misses	4	8

- Precision = $6 / (6 + 2) = .75$
- Recall = $6 / (6 + 4) = .60$

This dictionary is fairly precise, but it still misses many relevant speeches (recall is low)

- **Domain specificity**: a dictionary built for one corpus may not travel cleanly to another
- **Polysemy**: one word can have multiple meanings
- **False negatives**: relevant language may be missing from the term list
- **Language and translation**: concepts may not travel cleanly across languages / cultures

Choosing a dictionary strategy

- **Use off-the-shelf dictionaries:** only when the concept, language, and corpus are very close to the original dictionary
- **Avoid:** when meaning depends so strongly on context that counting terms will mislead
- Most projects sit between these extremes:
 - **adapt** an existing dictionary to the corpus
 - **build** a corpus-specific dictionary from theory, close reading, and validation

NB: adaptation starts from an existing instrument; building starts from a concept and a corpus.

Adapting a dictionary: fixing bills

Use adaptation when an existing dictionary is useful, but the corpus has its own language.

1. Start from the published dictionary and read the entries
2. Apply it to a sample of the corpus
3. Inspect KWIC examples for common hits
4. Remove or qualify terms that misfire
 - `bills` → `energy bills`, `household bills`, `council tax bills`
5. Add corpus-specific synonyms, variants, and multiword expressions
6. Validate the revised dictionary against human-coded examples

Building a dictionary: economic insecurity

Use building when the concept is domain-specific or no suitable dictionary exists.

1. Define the concept before looking for words
2. Close-read likely relevant and irrelevant documents
3. Identify candidate terms, phrases, wildcards, and exclusions
4. Search for differentially frequent words in “known” examples
5. Check each candidate in context and revise
6. Hand-code a validation sample and calculate agreement, precision, and recall

Source: adapted from Boumans & Trilling (2016) and Van Atteveldt et al. (2022)

Static embeddings for dictionary construction

Static embeddings can help expand a dictionary from a few seed terms.

- Start with theoretically motivated seeds: poverty, unemployment, inflation, bills
- Find words used in similar contexts in the corpus
- Inspect neighbours as **candidate terms**, not automatic additions
- Useful for historical or domain-specific language: MPs may discuss hardship without using our exact terms
- Still needs further validation (e.g., through KWIC-inspection)

LLM-assisted dictionary development

You are helping build a dictionary for economic insecurity in UK parliamentary speeches: hardship, unstable income, unaffordable living costs, debt, precarious work, and anxiety about basic needs.

Start from: poverty, unemployment, inflation, wages, rent, bills, debt.

Suggest additional terms for: income/wages, employment insecurity, cost of living, housing insecurity, debt, poverty/deprivation, welfare insecurity, and general hardship language.

Return a table: category | terms/phrases | why it fits | ambiguity or validation notes. Include UK policy terms, multiword phrases, wildcards, and likely false positives.

- Treat suggestions as hypotheses: inspect KWIC examples, revise the dictionary, and validate against human coding

Audit and validation checklist

Audit question	What to do
Does the term list match the concept?	read entries and define the construct
Which terms misfire?	inspect KWIC examples for common hits
Which terms have ambiguous meaning	Disambiguate meaning. Check grammatical
function of ambiguous words using POS tagging	
What language is missing?	close-read likely relevant speeches
Do patterns make sense?	compare rates by time, party, speaker, or agenda
Can humans reproduce it?	hand-code a validation sample

Report what was added, removed, and why.

Assessing content validity

- Content validation is mostly about the **word list**: add missing language, remove or qualify ambiguous terms, and document every change.

Check	In the economic insecurity example
Coverage	Are terms such as rent, arrears, food banks, Universal Credit, and fuel poverty included?
Ambiguity	Does bills mean household bills, energy bills, or parliamentary Bills?
Boundaries	Are broad terms such as cost, work, or support too general on their own?
Missing language	Do relevant speeches use phrases such as make ends meet, low pay, or temporary accommodation?

Assessing concept validity

- Concept validation is about the **measure**: do dictionary rates align with human judgments and substantive expectations?

Check	In the economic insecurity example
Human benchmark	Hand-code a sample of high, medium, and zero-score speeches
Error rates	Report precision, recall, and examples of false positives/negatives
Substantive contexts	Higher rates in welfare, poverty, housing, inflation, and energy debates
Time pattern	Plausible increases during cost-of-living or unemployment pressures
Negative cases	Low rates in procedural or unrelated debates

Takeaway

- Use dictionaries when you can defend the link between **words** and **concepts**
- Do not trust the count until you have inspected the context
- Embeddings and LLMs can help find candidate terms, but they do not validate the measure
- A good dictionary it is a **transparently documented** measurement instrument